Old Exam Questions
Computer Vision

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How this file is organized

This file is separate from the course summary and is meant as a clean practice set. Exact duplicates were removed. When essentially the same question appeared in multiple old exams, that is marked with *Asked multiple times.* inside the question box. The wording was lightly normalized where the scans were messy, but the mathematical content and answer choices were kept intact.

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Capturing Digital Images

Question 1: F-Number and Light

What does f/2.8 mean for a camera?

- ☐ Two times more light is captured than with f/4.
- ☐ The aperture diameter is 2.8 times the focal length.
- ☐ The focal length is 2.8 times the aperture diameter.
- ☐ Half the light is captured than with f/4.

Question 2: Lens Diameter

What is the diameter of an 80 mm focal length f/4 lens? (Answer in mm.)

Answer: _____

Question 3: Focal Length

Given f/8 and an aperture diameter of 16 mm, what is the focal length of a lens in mm? (Type in only the number.)

Answer: _____

Question 4: Shot Noise Distribution

Shot noise has what distribution?

- ☐ Gaussian
- ☐ Normal
- ☐ Poisson

Question 5: Demosaicing and Linearization

The relations between demosaicing and linearization of cameras are: (Multiple choices possible.)

- ☐ For computer vision applications, it makes sense to carry out linearization first and then demosaicing.
- ☐ For computer vision applications, it makes sense to carry out demosaicing first and then linearization.
- ☐ Bayer patterns and non-linear transfer functions mimic the human visual perception system.

Question 6: Barrel Distortion

Barrel distortion is present if . . .

- ☐ . . . the first coefficient (k_1) of the radial distortion polynomial is positive.
- ☐ . . . the first coefficient (k_1) of the radial distortion polynomial is negative.

- ☐ ...the pixel-distance to the principal point (d) is negative.
- ☐ ...the first coefficient (k_1) of the radial distortion polynomial is zero.

Question 7: LiDAR Distance Measurement

LiDAR determines distance by:

- ☐ (speed of light · time)/2
- ☐ (speed of sound · time)/2
- ☐ speed of sound · time
- ☐ speed of light · time

Question 8: Intrinsic Camera Parameters

What are intrinsic camera parameters? (Multiple choices possible.)

- ☐ Principal point
- ☐ Camera position
- ☐ Lens parameters
- ☐ Camera rotation
- ☐ Skew angle

Question 9: Homogeneous to Heterogeneous Coordinates

What needs to be done to convert homogeneous coordinates to heterogeneous coordinates?

- ☐ Transpose the coordinate vector.
- ☐ Extend the coordinate vector by a coefficient, being 1.
- ☐ Multiply the coordinate vector by a 3×3 matrix.
- ☐ Divide all coefficients by the last one.

Question 10: Learning-Based Camera Calibration

Learning-based approaches that perform camera calibration based on "in-the-wild" captured images: (Multiple choices possible.)

- ☐ Learn pixel-wise mapping using U-Nets.
- ☐ Learn from geometric representations, such as vanishing points.
- ☐ Regress intrinsic and extrinsic parameters using CNNs.

Question 11: Bayer Filter Color Distribution

The color filter of a sensor array (i.e., in a camera) is arranged in such a way that 50% red, 25% green, and 25% blue sub-pixels are imaged. (True/False)

- ☐ True
- ☐ False

Multi-View Geometry

Question 12: Mathematical Camera Models

What is true about mathematical camera models?

- ☐ Non-linear models must be used for thick camera lenses.
- ☐ Linear models can be used for thin camera lenses.

Question 13: Orthographic Projection and Vanishing Points

Orthographic projections have no vanishing points. (True/False)

- ☐ True
- ☐ False

Question 14: Essential and Fundamental Matrix

Which of the following statements are true? (Multiple choices possible.)

- ☐ The essential matrix can be applied to points on the physical image plane.
- ☐ The essential matrix is the multiplication of the inverse transpose of the first intrinsic matrix times the fundamental matrix times the inverse of the second intrinsic matrix.
- ☐ The fundamental matrix can be applied to points on the physical image plane.
- ☐ The fundamental matrix can be applied to points on the normalized image plane.
- ☐ The essential matrix can be applied to points on the normalized image plane.
- ☐ The fundamental matrix is the multiplication of the inverse transpose of the first intrinsic matrix times the essential matrix times the inverse of the second intrinsic matrix.

Question 15: Intrinsic Matrix Mapping

Multiplying the Intrinsic Matrix K ...

- ☐ ...maps coordinates from the physical image plane to the normalized image plane.
- ☐ ...maps coordinates from the normalized image plane to the physical image plane.
- ☐ ...maps coordinates between the physical image planes of two cameras.

Question 16: Epipolar Lines and Rectified Scanlines

Epipolar lines are rectified scanlines. (True/False)

- ☐ True
- ☐ False

Question 17: Epipolar Lines Count

Given a stereo-camera setup (cameras A and B): How many epipolar lines in camera B exist for one pixel in camera A? (Type in the number.)

Answer: _____

Question 18: Image Rectification in Stereo Vision

Which of the following are valid reasons for performing image rectification in stereo vision? (Multiple choices: select all that apply.)

- ☐ To make epipolar lines horizontal.
- ☐ To simplify correspondence search to 1D and in one direction.
- ☐ To avoid projection.
- ☐ To estimate camera intrinsic parameters.

Question 19: Structure-from-Motion: 3 Views, 3 Points

Can the structure-from-motion problem be solved for 3 views and 3 points? (True/False)

- ☐ True
- ☐ False

Question 20: Structure-from-Motion: Number of Equations

In a Structure-from-Motion problem with $m = 3$ camera views and $n = 20$ 3D points, how many equations are available if each point is visible in all views?

Answer: _____

Question 21: Structure-from-Motion: Number of Unknowns

How many unknowns do you have (in total) in the equation system of structure-from-motion with 25 points and 3 poses (considering all matrix coefficients to be unknown)? (Type in the number.)

Answer: _____

Question 22: Affine Shape

What is correct about affine shape? (Multiple choices possible.)

- ☐ It is correct only up to scale.
- ☐ You need at least 2 views of 4 points to reconstruct it.
- ☐ You need at least 2 views of 3 points to reconstruct it.
- ☐ It is correct only up to an affine transformation.
- ☐ You need at least 4 views of 3 points to reconstruct it.
- ☐ You need at least 3 views of 3 points to reconstruct it.

Question 23: Affine Structure-from-Motion: Unknown Coefficients

How many unknown coefficients do you have to determine in practice and in total for affine structure from motion using orthographic projection (with n points, m views)?

- ☐ $16m + 3n + 12$
- ☐ $27m + 3n + 18$

- ☐ $8m + 3n + 9$
- ☐ $8m + 3n$

Question 24: Projective Structure-from-Motion: Minimum Correspondences

Projective structure from motion (two-view case): Using the inequality $2nm > 11m + 3n - 15$, how many point correspondences are sufficient (minimum) to reconstruct two cameras and all the needed points in projective space?

Provide your answer in the format `NM`, where `N` is the number of points and `M` is the number of views.

Answer: _____

Question 25: Projection Matrix Correspondences

Without considering the individual intrinsic and extrinsic camera parameters, a 3D point $P = (x, y, z, 1)$ can be mapped to its 2D image coordinate $p = (u, v, 1)$ with a projection matrix M. How many corresponding pairs (P, p) are needed at least for a solution in M?

Answer: _____

Question 26: Trifocal Tensor Coefficients

How many coefficients does a trifocal tensor have in multiview vision?

- ☐ 27
- ☐ 9
- ☐ 3

Question 27: Affine vs. Euclidean Transformation

What is the difference between an affine and a Euclidean transformation?

- ☐ Rotation, translation, shear, scaling are all Euclidean transformations.
- ☐ A Euclidean transformation is a special form of an affine transformation.
- ☐ An affine transformation is a special form of a Euclidean transformation.
- ☐ Shear, scaling, warping are all affine transformations.

Question 28: Epipolar Constraints

Asked multiple times.

What is correct about epipolar constraints?

- ☐ The fundamental matrix can be applied to points on the normalized image plane, while the essential matrix can be applied to points on the physical image plane.
- ☐ The essential matrix can be applied to points on the normalized image plane, while the fundamental matrix can be applied to points on the physical image plane.

Question 29: Essential Matrix and Normalized Image Plane

Asked multiple times.

The essential matrix is applied to the normalized image plane! (True/False)

- ☐ True
- ☐ False

Question 30: Order Constraint in Stereo Matching

We cannot apply an order constraint when searching for matching features along scanlines (i.e., the feature order a,b,c,d,... along a scanline cannot be expected to be the same in each image). (True/False)

- ☐ True
- ☐ False

Digital Image Processing

Question 31: Convolution Properties

Which statement about convolution in image processing is correct?

- ☐ Convolution is a non-linear, shift-variant operation.
- ☐ Convolution requires the kernel to be the same size as the image.
- ☐ Convolution amplifies high-frequency components only.
- ☐ Convolution in the spatial domain corresponds to multiplication in the frequency domain.

Question 32: Convolution Rules

Given are two kernels H_1 and H_2, as well as an image I. Which of the following rules are correct? (Multiple choices possible.)

Note: k indicates a scaling factor and $*$ convolution.

- ☐ $H_1 * (H_2 * I) = (H_1 * H_2) * I$
- ☐ $I + (H_1 * H_2) = I * (H_1 + H_2)$
- ☐ $H_1 * I + H_2 * I = (H_1 + H_2) * I$
- ☐ $H_1 * (H_2 * I) = H_2 * (H_1 * I)$
- ☐ $(kH_1) * I = k(H_1 * I)$

Question 33: Fourier Domain Commutativity

If in spatial domain $f * g = g * f$, then in Fourier Domain $FG = G/F$. (True/False)

(f and g are image and kernel, $*$ is convolution, F and G are corresponding spectra, and $/$ is division.)

- ☐ True

☐ False

Question 34: Deconvolution in Frequency Domain

In theory, deconvolution in frequency domain is the FFT of the kernel divided by the FFT of the image. (True/False)

☐ True

☐ False

Question 35: 2D Poisson Equation

The 2D Poisson equation states that:

☐ The Laplacian of an image equals the curl of its gradient field.

☐ The Gaussian of an image equals the divergence of its gradient field.

☐ The Laplacian of an image equals the divergence of its gradient field.

☐ The Gaussian of an image equals the curl of its gradient field.

☐ The manipulated gradient field of an image can always be integrated.

Question 36: Laplacian Kernel Approximation

A Laplacian kernel can be approximated by ...

☐ ...subtracting an impulse (unit) from a Gaussian.

☐ ...subtracting two Gaussians with different sigma.

☐ ...subtracting a Gaussian from an impulse (unit).

Question 37: Filter Kernel Type

The filter kernel

$$\frac{1}{9}\begin{pmatrix} -1 & -1 & -1 \\ -1 & 17 & -1 \\ -1 & -1 & -1 \end{pmatrix}$$

is a ...

☐ edge filter

☐ smoothening filter

☐ sharpening filter

☐ Gaussian filter

Question 38: Hough Space Dimensionality for Ellipses

How high-dimensional is the Hough space (Hough transform) for segmenting 2D ellipses?

Answer: _____

Question 39: Hough Transform

The Hough space for finding lines in a 2D image is two-dimensional. Which statement about the Hough transform is correct?

- ☐ The Hough space for finding circles in a 2D image is two-dimensional.
- ☐ The Hough space for finding (infinitely large) planes in a 3D volume is two-dimensional.
- ☐ The Hough transform can only be applied to find lines.
- ☐ The Hough space for finding ellipsoids (3D version of ellipses) in a 3D volume is five-dimensional.

Question 40: ICP Complexity with K-D Trees

The complexity of ICP's (Iterative Closest Point Method) correspondence match, if no optimal data structures are used, is $O(N^2)$. What is its complexity roughly if k-d trees are used?

- ☐ $O(N \log N)$
- ☐ $O(N^3)$
- ☐ $O(N^2)$
- ☐ $O(N)$

Question 41: Eigenvectors and Variance

Eigenvectors represent directions along minimal variances in your data. (True/False)

- ☐ True
- ☐ False

Question 42: Graph-Based Segmentation: Cluster Count

In graph-based segmentation, an affinity matrix A of size $N \times N$ is constructed, where $a_{i,j}$ represents the similarity between pixels i and j. Suppose you use the clustering by graph eigenvectors method and compute the eigenvectors of A in each clustering round (iteration). Given:

- $N = 1000$ pixels
- The largest eigenvalue after the first round is $\lambda_1 = 85.3$
- The largest eigenvalue after the second round is $\lambda_2 = 42.7$
- The largest eigenvalue after the third round is $\lambda_3 = 18.4$

How many clusters will you obtain if you stop when the eigenvalue drops below 20.0?

Answer: _____

Question 43: Eigenfaces: Incorrect Statements

For Eigenfaces and Face Recognition, which of the following statements are **incorrect**? (Select all that apply.)

- ☐ The number of Eigenfaces (K) is chosen based on the largest eigenvalues, which corre-

spond to the directions of maximum variance.

- ☐ Face images are projected onto the Eigenface space by computing the dot product between the mean-subtracted image and each Eigenface.
- ☐ Eigenfaces are the eigenvectors of the covariance matrix computed from a set of face images.
- ☐ A new face image is represented as a weighted linear combination of the mean face and the Eigenfaces.
- ☐ Eigenface-based recognition is highly robust to extreme variations in lighting, pose, and facial expression between training and test images.

Question 44: Gray Code Hamming Distance

What is the Hamming distance of the Gray Code (often used for active range scanning / structured light), and why?

- ☐ It is 2 for better compression.
- ☐ It is 2 to detect errors in transmission.
- ☐ It is 1 for faster transmission compared to ordinary binary codes.
- ☐ It is 1 to detect errors in transmission.

Question 45: Gray Code Image Count

A Gray-Code projected with a resolution of 1024×1024 requires how many images to lead to a complete depth map?

- ☐ 30
- ☐ 15
- ☐ 10
- ☐ 18
- ☐ 20
- ☐ 9

Question 46: Helmholtz Reciprocity

The Helmholtz Reciprocity . . . (Multiple choices possible.)

- ☐ . . . can be implemented with the transpose light transport matrix.
- ☐ . . . can be implemented with the essential matrix.
- ☐ . . . states that the transport of light is geometrically invertible.
- ☐ . . . can be implemented with the inverse fundamental matrix.

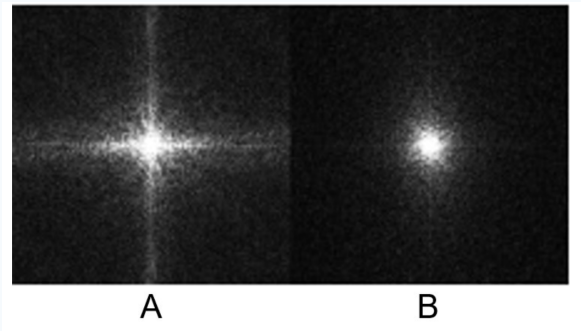
Question 47: Light Transport: Transpose Matrix

Light transport (dual photography / radiometric compensation): What is the result of $p = T'c$ (where T' is the transpose of T), assuming that c is a uniform white image?

- ☐ It computes an image that shows the scene from the perspective of the projector under a white illumination from the perspective of the camera.
- ☐ It results in a uniform white image from the perspective of the projector.
- ☐ It results in an image that shows the scene from the perspective of the camera, but where the scene depth appears reversed.

Question 48: Fourier Spectrum and Spatial Detail

Which of the two spectra belongs to the image with more spatial details? (Choose one.)



- ☐ A
- ☐ B

Question 49: Image Aggregation

What are examples for image aggregation? (Multiple choices possible.)

- ☐ Panorama imaging
- ☐ High-Dynamic Range (HDR) imaging
- ☐ Single-shot object detection

Question 50: Gradient Field Properties

Which of the following statements is correct? (Multiple choices possible.)

- ☐ The gradient field of an image has zero curl.
- ☐ The gradient field of an image is a conservative vector field.
- ☐ After processing the gradient field, the corresponding image can always be reconstructed by integration.
- ☐ The divergence of the gradient field is the Laplacian of the corresponding image.

Optical Flow

Question 51: Sparse Optical Flow Assumptions

What assumptions are made for sparse optical flow estimations (e.g., Lucas-Kanade)? (Multiple choices possible.)

- ☐ Spatial Coherence
- ☐ Color/Brightness Consistency
- ☐ Smooth Flow Fields

Question 52: Lucas-Kanade: Large Displacements

How do you compute optical flow that is larger than one pixel in the Lucas-Kanade optical flow approach?

- ☐ That is not possible. You need to apply a dense optical flow approach instead.
- ☐ You increase the considered neighborhood pixels to get more equations.
- ☐ You compute the optical flow at multiple resolutions.

Question 53: Lucas-Kanade vs. Horn-Schunck

Which statement best describes the main difference between the Lucas-Kanade and Horn-Schunck optical flow method?

- ☐ Lucas-Kanade estimates optical flow by solving a local least-squares problem, while Horn-Schunck estimates optical flow by minimizing a global energy function with a smoothness constraint.
- ☐ Horn-Schunck works only for small motion, while Lucas-Kanade handles large displacement without modification.
- ☐ Lucas-Kanade produces dense optical flow, while Horn-Schunck produces sparse optical flow only by default.
- ☐ Lucas-Kanade enforces global smoothness, while Horn-Schunck estimates motion independently per pixel.

Question 54: Stereo Optical Flow and Disparity

For a static rectified stereo-camera system and a static scene, the dense optical flow field of a recorded stereo pair correlates to a disparity map. (True/False)

- ☐ True
- ☐ False

Question 55: Deep Learning-Based Optical Flow

Which of the following statements about deep learning-based optical flow estimation are correct? (Multiple choices possible.)

- ☐ Encoder-decoder architectures (e.g., U-Net-like networks) are commonly used to produce dense optical flow fields.
- ☐ Convolutional neural networks for optical flow inference typically require explicit compu-

tation of image gradients.

- ☐ Deep learning methods estimate optical flow by learning a direct mapping from pairs of images to flow fields.

Deep Learning, Recognition, and Detection

Question 56: SIFT Scale Invariance

SIFT is not scale invariant. (True/False)

- ☐ True
- ☐ False

Question 57: Scale Invariance in CNNs

For model-based feature extraction, image pyramids are used to achieve scale invariance. What is used to achieve this in CNNs?

- ☐ Convolution
- ☐ Pooling and downsampling
- ☐ ReLU

Question 58: ResNet vs. Human Performance

ResNet can achieve a better top-5-error rate than humans. (True/False)

- ☐ True
- ☐ False

Question 59: ReLU Output

What is ReLU(-1234)?

Answer: _____

Question 60: ReLU Activation Purpose

In a Convolutional Neural Network (CNN), the Rectified Linear Unit (ReLU) activation function $f(x) = \max(0, x)$ is primarily used to:

- ☐ Introduce non-linearity and help mitigate the vanishing gradient problem.
- ☐ Normalize pixel values to a range between 0 and 1.
- ☐ Reduce the spatial dimensions of feature maps.
- ☐ Compute gradient orientations for feature descriptors.

Question 61: Max-Pooling Purpose

What is the reason for max-pooling in CNNs used for image classification?

- ☐ It averages feature responses and therefore it is more robust.
- ☐ It provides higher spatial accuracy for feature positions.
- ☐ It provides a large response regardless of the exact feature position.

Question 62: CNN Activation Layer Resolution

Given a Convolutional Neural Network using filter kernels of 7×7: What is the resolution of the 3rd activation layer if no pooling is applied, and if the input image resolution is 32×32? (Type in only one dimension, e.g., 32 for 32×32.)

Answer: _____

Question 63: Transposed Convolution Output Resolution

Your input image is 4×4 and your kernel is 3×3. What is the resolution of the output image after transposed convolution if your stride is 2 and padding is 1?

Note: if your output resolution is $A \times A$, type in only A.

Answer: _____

Question 64: CNN Architecture for Segmentation

What is a practical CNN architecture for segmentation tasks?

- ☐ A CNN with conv-layers of same resolution (without down-sampling).
- ☐ A conventional CNN where conv-layers are down-sampled.
- ☐ A CNN where conv-layers are down- and up-sampled inside the network.

Question 65: U-Net Skip Connections

In a U-Net architecture for semantic segmentation, what is/are the primary purpose(s) of skip connections between corresponding encoder and decoder layers? (Multiple choices possible.)

- ☐ Mid-level representations are often formed by clustering low-level features before classification.
- ☐ To preserve high-resolution spatial information lost during down-sampling.
- ☐ To reduce the number of trainable parameters.
- ☐ To introduce non-linearity into the decoder.
- ☐ To allow the network to bypass the bottleneck layer entirely.

Question 66: Instance Segmentation

Instance segmentation gives you a pixel-to-class assignment, but it is not possible to differentiate between different instances of the same class. (True/False)

- ☐ True
- ☐ False

Question 67: Scale-Depth Ambiguity

What is the standard option to overcome the scale-depth ambiguity problem when estimating depth-maps with CNNs?

- ☐ Assume smoothness in the gradients of the reconstructed depth map.
- ☐ Consider the global scene scale in the loss function.
- ☐ Segment the scene into different object-clusters first, and then estimate their depth values independently.
- ☐ Assume smoothness in surface normals of the reconstructed depth map.

Question 68: Scale-Invariant Loss for Depth Prediction

What is the primary purpose of using a scale-invariant loss function for monocular depth prediction?

- ☐ To improve the accuracy of surface normal prediction.
- ☐ To eliminate the need for ground truth depth data.
- ☐ To reduce computation time during training.
- ☐ To account for the fundamental scale-depth ambiguity in single-image depth estimation.

Question 69: Neural Networks for Depth Reconstruction

How can neural networks be used for depth reconstruction? (Multiple choices possible.)

- ☐ Fully connected layers can be used to learn the cost function.
- ☐ Autoencoders (U-Nets) can be used for end-to-end learning.
- ☐ U-Nets can be used to learn epipolar constraints.
- ☐ CNNs can be used for feature extraction.

Question 70: Cost-Volumes in PSMNet

Cost-Volumes in PSMNet (and other models) are used for . . . (Multiple choices possible.)

- ☐ . . . extracting features from multiple views at multiple scales with spatial pyramid pooling (SPP).
- ☐ . . . computing matching costs with a 3D CNN.
- ☐ . . . concatenating features from multiple views at all possible disparities.

Question 71: ResNet Layers for Depth from Defocus

Can individual conv-layers of ResNet be used for Depth from Defocus estimations? (Multiple choices possible.)

- ☐ Yes, because they provide feature saliency information that are proportional to focus.
- ☐ Yes, but only for uniform regions.
- ☐ No, because ResNet does not provide per-pixel information.
- ☐ Yes, but only for non-uniform regions.

- ☐ No, because ResNet is a model used for classification.

Question 72: Learned Filter Kernels in CNNs

For which conv layers of a CNN do you expect to find learned filter kernels that approximate Gaussians (assuming classical training databases, such as ImageNet)?

- ☐ Mid-level conv layers.
- ☐ High-level conv layers.
- ☐ Low-level conv layers.

Question 73: Vision Transformers

For Vision Transformers . . . (Multiple choices possible.)

- ☐ . . . only the decoder part is usually used.
- ☐ . . . linear projections of reshaped image patches + positional embeddings are usually used as input.
- ☐ . . . only the encoder part is usually used.
- ☐ . . . both encoder and decoder are usually used.

Question 74: Neural Radiance Fields: Inverse Problem

Neural Radiance Fields solve an inverse problem. (True/False)

- ☐ True
- ☐ False

Question 75: NeRF Architecture

The architecture behind classical Neural Radiance Fields (NeRFs) is . . .

- ☐ . . . a CNN.
- ☐ . . . a transformer.
- ☐ . . . a fully connected network.
- ☐ . . . a U-Net.

Question 76: NeRF Output

What does a Neural Radiance Field (NeRF) model output for a given 3D point and viewing direction?

- ☐ Vertex coordinates and texture.
- ☐ Depth and surface normal.
- ☐ Occupancy probability and gradient.
- ☐ Volume density and RGB color.

Question 77: NeRF with Binary Opacity

If you would condition the opacity solutions of Neural Radiance Fields to be binary, what kind of scenes would you still be able to represent with it? (Multiple choices possible.)

- ☐ Scenes with anisotropic reflections.
- ☐ Scenes with refractions.
- ☐ Scenes with specular reflections.
- ☐ Scenes with isotropic reflections.
- ☐ Scenes with diffuse reflections.

Question 78: Image–Text Correlation Model

What model can be used directly to learn correlations of images and text descriptions?

- ☐ Grad-CAM
- ☐ CLIP
- ☐ YOLO
- ☐ ResNET

Question 79: Large Multi-Modal Models

Which of the following statements about Large Multi-Modal Models (LMMs) are correct? (Multiple choices possible.)

- ☐ LMMs can be used for high-level vision tasks (e.g., reasoning) but may struggle with precise low-level tasks such as precise spatial estimations.
- ☐ LMMs suffer from poor generalization to novel visual concepts due to overfitting on language data.
- ☐ LMMs suffer from hallucination of visual content not present in images.
- ☐ LMMs can only perform tasks for which all input modalities are present at inference time.

Question 80: Non-Maximum Suppression in YOLO

Non-maximum suppression in YOLO . . .

- ☐ . . . removes bounding boxes that are too different.
- ☐ . . . removes bounding boxes that are too similar.

Question 81: Region Proposal Network in Faster R-CNN

What is the primary function of the Region Proposal Network (RPN) in Faster R-CNN?

- ☐ To compute the loss for objectness.
- ☐ To perform bounding box regression.
- ☐ To propose regions that likely contain objects.
- ☐ To classify objects into specific categories.

Question 82: Intersection over Union

The intersection over union of the two bounding boxes (red and blue) shown below is ... (Select all that apply.)

- ☐ < 0.5
- ☐ > 0.5
- ☐ > 1
- ☐ < 0
- ☐ < 1

Question 83: Precision

A detector produces the following results for a class: TP= 80, FP= 20, FN= 40. What is the precision?

Answer: _____

Question 84: Average Precision and Precision–Recall Curve

Let's assume your precision-recall curve is smooth; then an average precision of 0.5 indicates a relationship between recall and precision that is ...

- ☐ ... cubic.
- ☐ ... constant.
- ☐ ... quadratic.
- ☐ ... linear.

Question 85: K-Means Determinism

K-means clustering is deterministic. (True/False)

- ☐ True

☐ False

Question 86: Training Dataset Roles

Correlate the different parts of a training dataset with its functionality. Match each dataset split (left) to its role (right):

Test data _____

Training data _____

Validation data _____

Roles: *Determine final accuracy* *Train model parameters* *Tune hyperparameters*

Question 87: Event Cameras

Event cameras differ fundamentally from conventional frame-based cameras. Which of the following statements about event cameras are correct? (Multiple choices possible.)

- ☐ Event cameras inherently suffer from motion blur when observing fast-moving objects.
- ☐ Event cameras generally offer higher temporal resolution and lower latency than frame-based cameras.
- ☐ Event cameras asynchronously report changes in pixel intensity rather than capturing full image frames at fixed intervals.
- ☐ Event cameras require significantly more bandwidth than conventional cameras because they continuously stream all pixel values.
- ☐ Each event typically encodes the pixel location, timestamp, and the brightness change.

Question 88: CNN Trainable Parameters

A convolution layer is designed with input volume of $32 \times 32 \times 3$ and has 16 filters, each of size $3 \times 3 \times 3$. Assuming no bias terms are used, how many trainable parameters are in this convolutional layer?

Answer: _____

Question 89: Convolution Output Value

An image patch is given by

$$I = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

and the convolution kernel is

$$h = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix}.$$

What is the output value at the center pixel after convolution (assume valid convolution, no padding)?

Answer: _____

Question 90: Capturing Digital Images: Multiple Statements

Select all statements that are correct with respect to capturing digital images. (Multiple choices possible.)

- ☐ Color values in an RGB image measured through a Bayer filter are typically obtained by interpolation from neighboring subpixels.
- ☐ A higher signal-to-noise ratio generally results in more reliable intensity measurements.
- ☐ CMOS and CCD sensors measure the number of photons arriving at each pixel location over exposure time.
- ☐ Sensor noise has a stronger relative impact in dark image regions than in bright regions.
- ☐ In a Bayer filter pattern, each pixel directly measures all three color channels (R, G, and B).

Question 91: Autoencoder vs. U-Net Differences

Asked multiple times.

What are the differences between simple autoencoders and U-Nets used for segmentation?

- ☐ A simple autoencoder has an encoder and a decoder, while a U-Net has only an encoder.
- ☐ A U-Net has an encoder and a decoder, while a simple autoencoder has only an encoder.
- ☐ U-Nets have skip connections.

Question 92: Why U-Nets Excel at Segmentation

Asked multiple times.

U-Nets are better suited for segmentation tasks than unconnected autoencoders because ...

- ☐ ...of their additional pooling.
- ☐ ...of their additional convolutions.
- ☐ ...of their additional skip-connections.
- ☐ ...of their additional transposed convolutions.

Question 93: Traditional Model-Based Classification Pipeline

In a traditional model-based classification pipeline (e.g., using handcrafted features like SIFT), which of the following statements are true?

- ☐ It typically involves steps such as feature detection, descriptor extraction, and clustering.
- ☐ Feature detectors are manually designed and remain fixed (e.g., edge, corner, and blob detectors).
- ☐ Mid-level representations are often formed by clustering low-level features before classification.
- ☐ The feature extraction kernels are learned automatically from labeled training data.
- ☐ It requires no preprocessing or invariance handling (e.g., scale or rotation invariance).

Answer Key

Q1. **a, c**

Q2. 20 mm

Q3. 128 mm

Q4. **c**

Q5. **a, c**

Q6. **a**

Q7. **a**

Q8. **a, c, e**

Q9. **d**

Q10. **a, b, c**

Q11. **b** (False)

Q12. **a, b**

Q13. **a** (True)

Q14. **c, e, f**

Q15. **b**

Q16. **b** (False)

Q17. 1

Q18. **a, b**

Q19. **b** (False)

Q20. 120

Q21. 111

Q22. **b, d**

Q23. **c**

Q24. 72 (7 points, 2 views)

Q25. 6

Q26. **a** (27)

Q27. **b**

Q28. **b**

Q29. **a** (True)

Q30. **a** (True)

Q31. **d**

Q32. **a, c, d, e**

Q33. **b** (False)

Q34. **b** (False)

Q35. **c**

Q36. **c**

Q37. **c**

Q38. 5

Q39. **d**

Q40. **a**

Q41. **b** (False)

Q42. 2

Q43. **e**

Q44. **d**

Q45. **e** (20)

Q46. **a, c**

Q47. **a**

Q48. **a**

Q49. **a, b**

Q50. **a, b, d**

Q51. **a, b**

Q52. **c**

Q53. **a**

Q54. **a** (True)

Q55. **a, c**

Q56. **b** (False)

Q57. **b**

Q58. **a** (True)

Q59. 0

Q60. **a**

Q61. **c**

Q62. 14

Q63. 7

Q64. **c**

Q65. **b**

Q66. **b** (False)

Q67. **b**

Q68. **d**

Q69. **a, b, d**

Q70. **b, c**

Q71. **a, d**

Q72. **c**

Q73. **b, c**

Q74. **a** (True)

Q75. **c**

Q76. **d**

Q77. **d, e**

Q78. **b**

Q79. **a, c**

Q80. **b**

Q81. **c**

Q82. **b, e**

Q83. 0.8

Q84. **d**

Q85. **b** (False)

Q86. Test data → Determine final accuracy; Training data → Train model parameters; Validation data → Tune hyperparameters

Q87. **b, c, e**

Q88. 432

Q89. 0

Q90. **a, b, c, d**

Q91. **c**

Q92. **c**

Q93. **a, b, c**